CNHR Dashboard

User Guide

Comprehensive Reference for the Cost-Normalised Hourly Rate Analytical Dashboard

Model Version	CNHR v1.0
Data Period	December 2025 – February 2026
Total Trips	427 across 10 Uber weeks
Fixed Cost (C_w)	£430 per week
Target Rate (r*)	£15 per hour net
Reference	CNHR_Model_Formal_Specification.pdf
Date Issued	12 February 2026

Executive Summary

This guide explains the CNHR Dashboard—a visual tool that tracks the real profitability of Private Hire Vehicle (PHV) operations after accounting for all fixed weekly costs. The core insight: gross earnings per trip are misleading. Only by subtracting the full weekly cost base and dividing by total engaged hours do you see your true hourly take-home rate.

Key Findings at a Glance

Aggregate CNHR	£8.48/hr — RECOVERY phase (positive but below £15/hr target)
10-Week Earnings	£5,840 from 427 trips over 181.6 engaged hours
Gross Hourly Rate (ρ)	£32.15/hr — strong per-trip performance
Weeks Reaching Target	3 of 10 (30%) — all exceeded 26 engaged hours
Weeks Covering Costs	6 of 10 (60%) — earnings exceeded £430 in these weeks
Critical Threshold	~13.4h to break even; ~25h to reach target rate at aggregate ρ
Binding Constraint	**Hours, not trip quality. Every deficit week had fewer than 15 engaged hours.**

Bottom line: individual trips are profitable (median £32.70/hr gross). The gap between gross earnings and net take-home is entirely driven by the £430/week fixed cost base spread across too few hours. Weeks with 26+ engaged hours consistently reach or approach the £15/hr CNHR target.

Understanding the CNHR Model

The Cost-Normalised Hourly Rate (CNHR) model, formally specified in CNHR_Model_Formal_Specification.pdf, measures the true profitability of PHV operations by accounting for fixed weekly costs that standard gross earnings ignore.

The Core Equation

$$r_t = (E_t - Cw) / T_t$$

r_t	CNHR — your net hourly rate after all costs. This is the primary metric.
E_t	Cumulative gross earnings for the week (trip payments, holiday pay, Central London fees, tips).
Cw	Weekly fixed cost = £430. Includes vehicle rental (£220), insurance, maintenance, charging, and all other operating expenses.
T_t	Total engaged hours for the week (time actively transporting passengers, as reported by Uber).
r*	Target CNHR = £15/hr. The minimum net rate for the operation to be strategically viable.

Why CNHR Matters

A driver earning £30/hr gross may appear highly profitable. But if that driver works only 12 hours in a week, gross earnings total £360—less than the £430 cost base. The CNHR for that week would be (360 − 430) / 12 = −£5.83/hr. The driver is paying £5.83 for every hour worked. Gross rate hides this reality; CNHR exposes it.

The Three Phases

Phase	Condition	Meaning
TARGET	$r_t \geq$ £15/hr	All costs recovered AND earning above target. Optimal zone.
RECOVERY	$0 \leq r_t <$ £15/hr	Costs recovered but earning below target. Profitable but sub-optimal.
DEFICIT	$r_t < 0$	Costs NOT recovered. Net loss for the week. Every hour driven deepens the loss.

These phases are defined formally in Definition 2.4 of the CNHR_Model_Formal_Specification.pdf and are colour-coded throughout the dashboard: green for TARGET, amber for RECOVERY, red for DEFICIT.

Two Critical Inflection Points

Break-Even (★ BE): The exact trip at which cumulative earnings first equal or exceed Cw (£430). Before this point, r_t is negative (DEFICIT). After it, r_t turns positive (RECOVERY or TARGET). On the dashboard, this is marked with a green vertical line (★ BE).

Target Achievement (★ r*): The exact trip at which r□ first reaches or exceeds £15/hr. This requires not only recovering all costs but accumulating enough surplus that the net rate clears the target. Marked with a purple vertical line (★ r*). Per the formal specification (Theorem 3.2), achieving r* requires T□ ≥ Cw / (ρ − r*), where ρ is the aggregate gross rate.

Dashboard Sections Guide

Aggregate Statistics Bar

The six metric boxes across the top of the dashboard provide the 10-week aggregate view. These are computed from portal trip-level data (which excludes promotional bonuses and NLW holiday pay top-ups, so statement totals may be higher).

Metric	Value	Explanation
Total Earnings	£5,840	Sum of all trip-level earnings across all 10 weeks from portal data.
Total Hours	182h	Sum of engaged trip durations. 427 completed trips.
Aggregate ρ	£32.15/hr	Gross hourly rate = Total Earnings / Total Hours. Ignores costs entirely.
Aggregate CNHR	**£8.48/hr**	Net hourly rate = (Total Earnings − £430 × 10 weeks) / Total Hours.
Weeks at r*	3/10	Number of weeks where final r□ ≥ £15/hr.
Weeks at BE	6/10	Number of weeks where total earnings ≥ £430.

Overview Charts

Weekly Earnings vs Break-Even (left chart): A bar chart showing each week's total portal earnings. Bars are colour-coded by phase (green/amber/red). The horizontal red dashed line at £430 marks the break-even threshold. Bars above this line represent weeks where costs were covered.

CNHR Trajectory & Hours (right chart): A combined chart. Blue bars show engaged hours per week. The amber line with dots traces the final r□ for each week. The purple dashed line at £15/hr marks the target. This chart reveals the correlation between hours worked and CNHR achieved.

Week Selector Cards

The horizontally scrollable row of week cards enables selection of any individual week for detailed intra-week analysis. Each card displays:

Week date range (Uber weeks run Monday 04:00 to the following Monday 03:59), trip count, total portal earnings, engaged hours, gross rate (ρ), final CNHR (r□) in phase colour, and the trip numbers at which break-even and target were reached (or "No BE" / "No r*" if not achieved).

Tapping or clicking a card loads that week's data into the detail panel below.

Intra-Week Detail Panel

This is the core analytical view. It shows how profitability evolves trip by trip within the selected week.

Chart Modes

Three toggle buttons switch the chart view:

Cumulative £: Shows the running total of earnings climbing from zero toward and beyond the £430 break-even line (red dashed). The blue area fill under the line helps visualise the earnings accumulation. Green vertical line marks exact break-even; purple marks target achievement.

CNHR r□: Shows the running CNHR value converging from deeply negative (early trips bear the full cost burden) toward zero and potentially above £15/hr. Two lines: solid amber = r□ (net rate), dashed cyan = ρ (gross rate). The gap between them represents the cost drag. The CNHR curve's characteristic shape—starting extremely negative then rapidly climbing and gradually flattening—reflects the mathematical property that fixed costs are amortised over increasing hours (see Theorem 3.1 of the formal specification).

Trip ρ: Bar chart of each individual trip's gross hourly rate. Bars are coloured: green (≥£40/hr), blue (≥£25/hr), amber (≥£15/hr), red (<£15/hr). This reveals trip-level quality variation and helps identify which trip types or time slots yield the strongest rates.

Inflection Point Detail Boxes

Below the chart, two side-by-side panels provide precise data about each critical inflection point:

Break-Even Box (green border): When reached, shows the exact trip number, day and time, cumulative earnings and hours at that moment, the pickup and drop-off locations of the break-even trip, and how many remaining trips contributed beyond break-even. When not reached, shows the shortfall amount and how many additional hours at the week's gross rate would have been needed.

Target Rate Box (purple border): Same structure as break-even but for the r* = £15/hr threshold. Shows the trip at which net earnings per hour first clears target, or the additional hours required at current ρ if not achieved.

Trip Table

The scrollable table at the bottom provides the complete trip-by-trip dataset for the selected week. Each row represents one completed trip.

Column	Description
#	Sequential trip number within the week. ★ = break-even trip. ◆ = target achievement trip.
Day / Time	Day of week and start time of the trip.
Svc	Uber service type: UberX, Electric, Comfort, or UberX Priority.
Earn	Total earnings for this individual trip (trip payment + holiday pay + Central London fee + tip).
Dur	Trip duration in minutes (engaged time transporting the passenger).
ρ trip	This trip's gross hourly rate: Earn / (Dur in hours). Colour-coded by quality.
Cum £	Cumulative earnings from trip #1 through this trip.
Cum H	Cumulative engaged hours from trip #1 through this trip.

ρ avg	Running average gross rate: Cum £ / Cum H.
r□	Running CNHR: (Cum £ − £430) / Cum H. The key metric. Colour-coded by phase.
Rec %	Cost recovery percentage: (Cum £ / £430) × 100. Reaches 100% at break-even.
Phase	Current phase badge: DEFICIT (red), RECOVERY (amber), or TARGET (green).

Rows are highlighted with green background at the break-even trip and purple background at the target achievement trip for quick visual identification.

Week-by-Week Results

The following table summarises all 10 Uber weeks. Note that portal trip-level data excludes promotional bonuses, so statement totals are higher for weeks where promotions were received (particularly 08–15 Dec which included £400 in promos).

Week	Trips	Earn	Hours	ρ	CNHR	BE	r*	Phase
01 Dec–08 Dec	13	£125.60	4.2h	£30.17	**£-73.11**	—	—	**DEFICIT**
08 Dec–15 Dec	54	£884.01	26.3h	£33.66	**£17.29**	#21	#47	**TARGET**
15 Dec–22 Dec	47	£677.34	14.8h	£45.67	**£16.68**	#27	#45	**TARGET**
29 Dec–05 Jan	34	£400.48	10.8h	£36.94	**£-2.72**	—	—	**DEFICIT**
05 Jan–12 Jan	29	£423.21	14.3h	£29.53	**£-0.47**	—	—	**DEFICIT**
12 Jan–19 Jan	53	£674.89	22.9h	£29.51	**£10.71**	#34	—	**RECOVERY**
19 Jan–26 Jan	27	£373.23	13.1h	£28.51	**£-4.34**	—	—	**DEFICIT**
26 Jan–02 Feb	50	£760.98	26.2h	£29.03	**£12.63**	#32	—	**RECOVERY**
02 Feb–09 Feb	71	£879.05	28.4h	£30.95	**£15.81**	#31	#67	**TARGET**
09 Feb–16 Feb	49	£641.41	20.6h	£31.09	**£10.25**	#34	—	**RECOVERY**

Data Completeness

Portal daily JSON files were not uploaded for 21–30 December. This means the 15–22 Dec week has partial data (missing Dec 21), the 22–29 Dec week is entirely absent from the dashboard, and the 29 Dec–05 Jan week is missing Dec 29–30. Statement totals remain authoritative for these periods. For weeks with complete portal data (05 Jan onward), trip counts reconcile exactly or within ±1 trip due to the Monday 04:00 boundary assignment.

Uber Week Boundary

Uber's operational week runs from Monday at 04:00 to the following Monday at 03:59. A trip starting at Sunday 23:45 belongs to the current week. A trip starting at Monday 03:30 also belongs to the current (ending) week. A trip starting at Monday 04:01 belongs to the new week. The dashboard assigns trips by start time to match this boundary.

Interpreting the Charts

Reading the Cumulative Earnings Chart

The S-shaped earnings curve reveals the rhythm of a working week. Flat segments indicate inactive periods (daytime rest, days off). Steep segments indicate high-earning sessions. The critical question is whether the curve crosses the red £430 line, and if so, when.

Early break-even (before trip #25) indicates either high earnings per trip, high session density, or both. These weeks have the most time in the TARGET or RECOVERY zone.

Late or no break-even indicates insufficient hours or earnings. Every deficit week in this dataset had fewer than 15 engaged hours—well below the ~13.4h theoretical minimum needed at the aggregate ρ of £32.15/hr.

Reading the CNHR Convergence Chart

The amber r□ line always starts at an extreme negative value after the first trip (the full £430 cost is divided by a fraction of an hour). This is expected and not cause for concern. The line then rises rapidly as more hours accumulate, because the fixed cost is being spread across increasing time.

The mathematical convergence (Theorem 3.1 in the formal specification) shows that as T□ grows, r□ approaches ρ asymptotically. The gap (ρ − r□) = Cw / T□, which shrinks hyperbolically with hours. This means each additional hour of work has diminishing marginal impact on CNHR—the first 10 hours make the biggest difference.

The dashed cyan ρ line (gross rate) provides the ceiling that r□ can never reach. It tends to be more stable across the week because it is unaffected by costs. When ρ is volatile, it indicates inconsistent trip quality.

Reading the Trip Rate Bars

This view exposes per-trip variance. In a healthy week, most bars should be blue (≥£25/hr) or green (≥£40/hr). Red bars (<£15/hr) indicate trips that individually earn less than your net target rate—though even these contribute to cost recovery when the weekly aggregate ρ exceeds the break-even threshold.

Clusters of red bars may indicate specific time periods, routes, or service types that consistently underperform and could be candidates for avoidance in Sentinel's trip acceptance logic.

Glossary of Terms

Term	Definition
CNHR (r_n)	Cost-Normalised Hourly Rate. Net earnings per hour after subtracting all fixed weekly costs. The primary profitability metric.
Gross Rate (ρ)	Total earnings divided by total engaged hours, ignoring all costs. Shows per-trip earning power.
Cw (Weekly Cost)	Fixed weekly operating cost = £430. Includes vehicle rental (£220/week), insurance, charging, maintenance, phone, and other overheads.
r* (Target Rate)	The minimum acceptable net hourly rate = £15/hr. Represents the strategic viability threshold.
E_n (Earnings)	Cumulative gross earnings at trip n. Includes trip payments, per-trip holiday pay, Central London fees, and tips.
T_n (Hours)	Cumulative engaged hours at trip n. Time actively transporting passengers as reported by Uber.
Break-Even	The point where E_n = Cw. After this, every additional pound earned contributes to positive CNHR.
DEFICIT Phase	r_n < 0. Weekly costs exceed earnings. Continuing to drive reduces the deficit but the driver is still net negative.
RECOVERY Phase	0 ≤ r_n < r*. Costs recovered but net rate below target. Profitable but not meeting strategic objectives.
TARGET Phase	r_n ≥ r*. Costs fully recovered and net rate at or above target. The operational sweet spot.
Engaged Hours	Uber's reported time spent actively transporting passengers. Excludes waiting, repositioning, and deadhead miles.
Portal Data	Trip-level data from Uber's driver portal (daily JSON exports). Excludes weekly promotions and NLW top-ups.
Statement Data	Uber's official weekly NLW statements (txt). Includes promotions, holiday pay adjustments, and guaranteed minimums. Authoritative for totals.
Uber Week	Monday 04:00 to the following Monday 03:59. All weekly metrics use this boundary.
Sentinel	The intelligent trip advisory system under development that will use CNHR principles to recommend trip acceptance/rejection in real time.
Recovery %	Percentage of weekly costs recovered: (E_n / Cw) × 100. Reaches 100% at break-even.

Strategic Implications

Hours Are the Binding Constraint

The data unambiguously shows that the aggregate gross rate (£32.15/hr) is strong—more than double the target r* of £15/hr. The problem is not earning power per trip; it is total engaged hours per week. Every week that achieved TARGET phase had 26+ engaged hours. Every DEFICIT week had fewer than 15 hours.

At the observed ρ = £32.15/hr, the theoretical thresholds from the formal specification (Theorem 3.2) are: **break-even at 13.4 hours** and **target rate at 25.1 hours.**

Cost Structure Impact

The formal specification's sensitivity analysis (Section 5) demonstrates that reducing Cw from £430 (rental) to £210 (estimated ownership cost) would transform the aggregate CNHR from £8.48/hr to approximately £20.59/hr—well above target. Under the ownership model, 8 of 10 weeks would achieve TARGET phase versus only 3 under rental. This represents the single largest structural lever available.

Implications for Sentinel

The trip acceptance algorithm should prioritise hours maximisation over per-trip rate optimisation. Rejecting a £25/hr trip to wait for a £35/hr trip costs idle time that directly reduces weekly hours. The exception is trips with ρ below £20/hr, which contribute minimally to cost recovery and may indicate unfavourable repositioning. Sentinel should target a minimum trip ρ of £20/hr while maximising accepted trip volume.

Reference Document: CNHR_Model_Formal_Specification.pdf contains the complete mathematical framework including all definitions, theorems, proofs, and sensitivity analyses referenced in this guide. The formal specification and this dashboard together provide the analytical foundation for Sentinel's trip advisory logic.